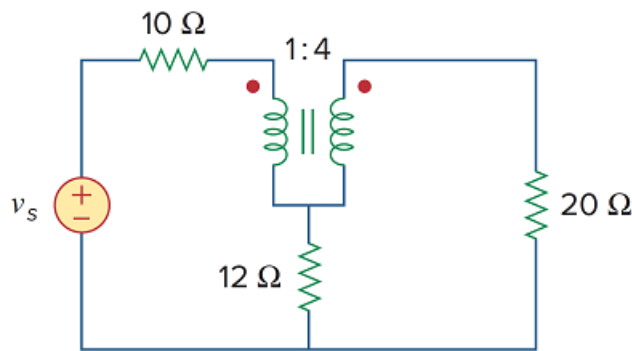


Problem

In the circuit of Fig. 13.124, let $v_s = 165 \sin(1,000t)$ V. Find the average power delivered to each resistor.

Figure 13.124



Step-by-step solution

Step 1 of 11

Refer to Figure 13.124 from the textbook.

Assume the voltage across the primary of the transformer be V_1 and voltage across secondary winding to be V_2 . Assume the direction of the loop currents to be I_1 and I_2 as shown in Figure 1.

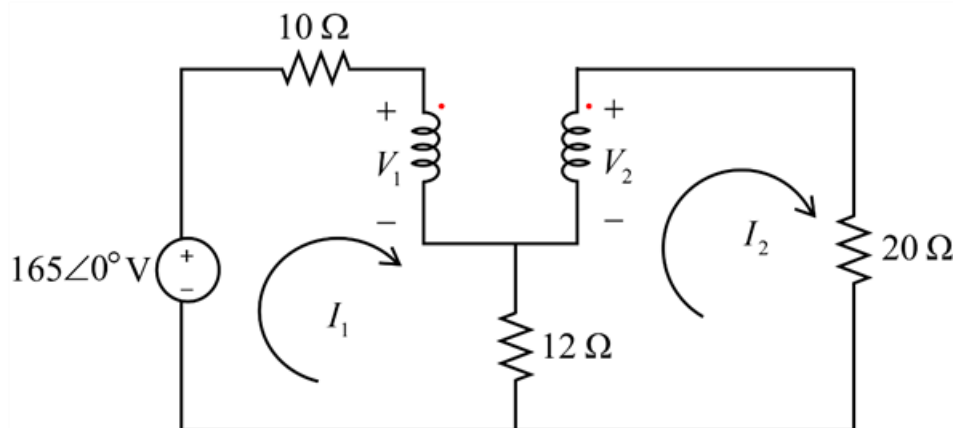


Figure 1

Step 2 of 11

For a transformer,

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

Here,

N_1 are the number of turns in primary winding of the transformer,

N_2 are the number of turns in secondary winding of the transformer.

Substitute $\frac{1}{4}$ for $\frac{N_1}{N_2}$.

$$\frac{V_1}{V_2} = \frac{1}{4}$$

$$V_2 = 4V_1$$

Step 3 of 11

Write the ratio of primary current to the secondary current for the transformer.

$$\frac{I_2}{I_1} = \frac{N_1}{N_2}$$

Substitute $\frac{1}{4}$ for $\frac{N_1}{N_2}$.

$$\frac{I_2}{I_1} = \frac{1}{4}$$

$$I_1 = 4I_2 \dots\dots (1)$$

Step 4 of 11

Apply the Kirchhoff's Voltage Law in Loop 1.

$$165\angle 0^\circ = 10I_1 + V_1 + 12(I_1 - I_2)$$

$$165\angle 0^\circ = 22I_1 - 12I_2 + V_1 \dots\dots (2)$$

Step 5 of 11

Apply the Kirchhoff's Voltage Law in Loop 2.

$$V_2 = 20I_2 + 12(I_2 - I_1)$$

$$V_2 = 32I_2 - 12I_1$$

Substitute $4V_1$ for V_2 .

$$4V_1 = 32I_2 - 12I_1$$

$$V_1 = \frac{32I_2 - 12I_1}{4}$$

$$V_1 = 8I_2 - 3I_1$$

Step 6 of 11

Substitute $8I_2 - 3I_1$ for V_1 in equation (2).

$$165\angle 0^\circ = 22I_1 - 12I_2 + 8I_2 - 3I_1$$

$$165\angle 0^\circ = 19I_1 - 4I_2$$

Substitute $4I_2$ for I_1 .

$$165\angle 0^\circ = 19(4I_2) - 4I_2$$

$$165\angle 0^\circ = 72I_2$$

$$I_2 = \frac{165\angle 0^\circ}{72}$$

$$I_2 = 2.292\angle 0^\circ \text{ A}$$

Step 7 of 11

Calculate the current I_1 .

Substitute 2.292 A for I_2 in equation (1).

$$I_1 = 4(2.292) \\ 9.167 \text{ A}$$

Step 8 of 11

Write the expression for the average power delivered to a resistor, R .

$$p = \frac{1}{2} I^2 R \dots\dots (3)$$

Step 9 of 11

Calculate the average power delivered to the 10Ω resistor, p_1 .

$$p_1 = \frac{1}{2} I_1^2 R$$

Substitute 9.167 A for I_1 and 10Ω for R .

$$p_1 = \frac{1}{2} (9.167)^2 (10) \\ = \frac{1}{2} (84.028)(10) \\ = 420.13 \text{ W}$$

Step 10 of 11

Calculate the average power delivered to the $12\ \Omega$ resistor, p_2 .

$$p_2 = \frac{1}{2}(I_1 - I_2)^2 R$$

Substitute 9.167 A for I_1 , 2.292 A for I_2 and $12\ \Omega$ for R .

$$\begin{aligned} p_2 &= \frac{1}{2}(9.167 - 2.292)^2 (12) \\ &= \frac{1}{2}(6.875)^2 (12) \\ &= \frac{1}{2}(47.26)(12) \\ &= 283.59\text{ W} \end{aligned}$$

Step 11 of 11

Calculate the average power delivered to the $20\ \Omega$ resistor, p_3 .

$$p_3 = \frac{1}{2}I_2^2 R$$

Substitute 2.292 A for I_2 and $20\ \Omega$ for R .

$$\begin{aligned} p_3 &= \frac{1}{2}(2.292)^2 (20) \\ &= \frac{1}{2}(5.25)^2 (20) \\ &= \frac{1}{2}(27.49)(20) \\ &= 52.52\text{ W} \end{aligned}$$

Therefore, the average power delivered to $10\ \Omega$ the resistor, p_1 is 420.13 W , the average power delivered to $12\ \Omega$ the resistor, p_2 is 283.59 W and the average power delivered to $20\ \Omega$ the resistor, p_3 is 52.52 W .